

[DC] A Mobile Intervention to Promote Social Skills in Children with Autism Spectrum Disorder Using AR Face Masks

Hiroshika Premarathne*

Victoria University of Wellington, New Zealand
University of Kelaniya, Sri Lanka

ABSTRACT

Autism Spectrum Disorder is a lifelong neuro-developmental disorder characterized by several behaviors including the deficits in emotion recognition and face-directed eye gaze. Although, behavioral therapists are there to help children with ASD to improve the skills, due to both high demand for services and lack of resources, there is a need for providing alternative interventions to therapy sessions. My PhD aims to develop an Augmented Reality (AR) based mobile application to provide an alternative method and assist children with ASD in identifying emotions and communicating with their close family members using face-directed eye gaze.

Index Terms: Augmented reality—AR face masks—Social stories—Video modeling—Co-designing—Discrete trial training—Emotion recognition—User-centered Design

1 INTRODUCTION

The significant and lifelong challenges faced by children with ASD, in particular with regards to social skills training, result in a high demand of services and professionals that can support them. But, children are often unable to start their therapy sessions, as there are long wait lists for admittance due to the unavailability of behavioral therapists. In response to this, there is a significant body of work aiming to address these challenges and support the therapy of children with ASD in emotion recognition and face directed eye gaze. There are a plethora of technologies used to enhance the emotion recognition and face-directed eye gaze of children with ASD such as virtual reality (VR) and wearable technologies. While these have shown some promising results, they each have their own disadvantages. VR creates an isolated environment [8]. Finding relevance of a social skill in a different social context is difficult for autistic children, if they are trained in a different setup and there is an open question whether the skills developed in the virtual world are actually valid in the real world. [9]. And wearable technologies have been found to be invasive [10]. In contrast, AR does not suffer from the above shortcomings and moreover, it has been found to engage positively children with autism [7] thanks to its attractiveness and provided fun [1].

A lot of researchers have conducted studies using AR technologies such as with smartglasses and different setups in support of social skill development of children with ASD. But only a handful studies have been conducted with AR based mobile applications. A mobile application called MOSOCO [6] has used AR to support the social skill development of students with ASD. to provide visual support for children to practise in real-life situations outside the classroom. It is encouraging the students to make eye contact, maintain appropriate spatial boundaries, reply to conversation initiators, share interests with partners, disengage appropriately at the

end of an interaction, and identify potential communication partners. Similarly, Cunha et al. [4] have developed a mobile-based story book introducing a character called Tobias who is behaving in different situations and has to attempt a small quiz at the end of each chapter to choose the correct emotion in a given situation. But, the system was not tested for usability and effectiveness with children with ASD, and there was no function to measure their progress in terms of the emotion recognition component [2, 4]. In my PhD, I am proposing a more efficient, effective and engaging solution to overcome these problems.

2 RESEARCH QUESTIONS AND CONTRIBUTIONS

The research question is; How an AR-based mobile application can be used as an alternative intervention method and a communication method in social conversations of autistic children with their close family members, by maintaining face directed eye-gaze and emotion recognition?

A series of studies will be conducted based on these sub questions, to answer the research question; **RQ1:** Do the AR face masks foster the maintenance of face-directed eye gaze during the conversation? **RQ2:** Do the AR face masks enhance the engagement of the conversation? **RQ3:** What features can be altered to create more customized AR face masks for children with ASD? **RQ4:** How AR storytelling can be used to identify the emotions of the characters in a specific situation and is this skill transferable to similar real-world situations?

Two contributions will be made through this PhD work. First, an AR mobile application which will function as an alternative therapeutic intervention leveraging on Discrete Trial Training (DTT) and incidental teaching (IT). Second, a co-design process will be followed throughout the development of the AR application which, together with an iterative evaluation approach, will provide detailed insights of the children's requirements, and inform the effectiveness and level of user satisfaction of using AR masks and video modelling in this context.

3 DESIGN AND IMPLEMENTATION

This research will be conducted using a user-centered design approach [3] and by incorporating co-design [11] to offer the best possible user experience for children with ASD. The developing prototypes will be refined iteratively based on feedback obtained by children with ASD. The direct involvement of children with ASD in the research and development process, will provide significant knowledge towards their requirements, appropriate implementation solutions and context of use.

The mobile application is composed of two main components to provide children with ASD, an alternative communication method using face directed eye gaze and emotion recognition to have conversations with their non-autistic parents or caregivers. The face directed eye gaze component is using AR based masks on the parent's face which will be implemented at the parent's end after the video call is initiated. Unity face AR plugin will be used to develop the AR based masks. For the emotion recognition component, a set of videos will be displayed and set of questions and answers will

*e-mail: hiroshika.hinduramage@vuw.ac.nz, hiroshika@kln.ac.lk

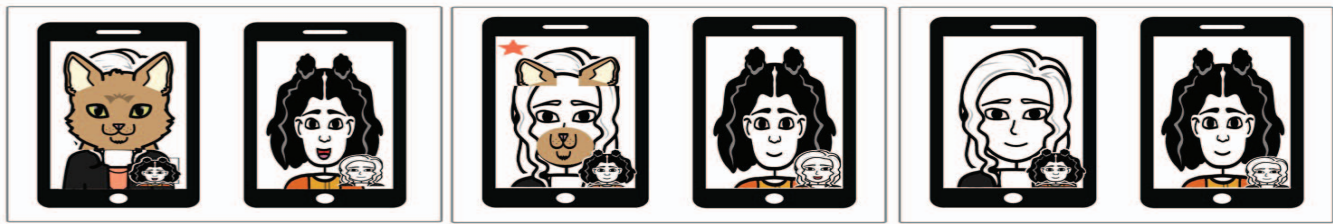


Figure 1: The form of DTT application in the component of face directed eye gaze

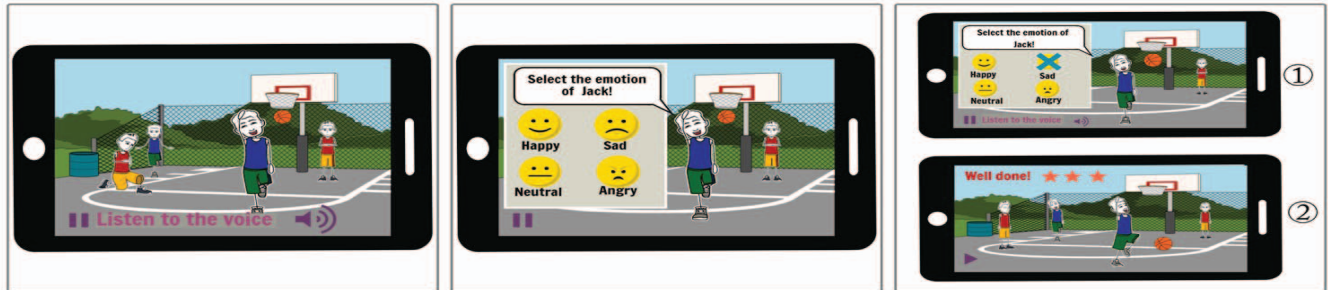


Figure 2: The form of DTT application in the component of emotion recognition

be utilized. They will be stored on a server. Progress data such as rewards and achieved milestones will be stored in the local SQLite DB of the child's end.

4 CURRENT RESEARCH AND FUTURE WORK

The first study of this PhD project aimed to understand the existing methodologies and tools used by professional therapists to support the social skills training of children with ASD. To do so, semi-structured interviews were conducted with them. Currently, we are at the final stages of analysing the interview data and three themes have been developed using Thematic Analysis: Therapeutic approaches in practice; Use of technology in behavioral therapy; and Effects of reinforcement. The initial design of the mobile application has been designed applying the outcomes of the interviews and existing literature in the areas of autism, AR, DTT and IT.

The frequency and duration of face-directed eye-gaze is expected to be assisted through this face directed eye gaze component. This is to make a video call to a parent and have a friendly conversation. When the conversation starts, the parent's face will be overlaid with an AR face mask of some character which is the discriminative stimulus and also it works as the conditioned transitive motivating operation. When the child looks at the face area for a pre-decided duration or a frequency, it will start to fade out until the child faces the real person at the end which is called as prompt fading. When the child looks at the parent, the reinforcement will be provided to motivate the child. Fig. 1 shows an example view of face directed eye-gaze component.

In the emotion recognition component, video modeling and storytelling will be used as the main intervention method along with game integration. While the modelled social story video is playing, it will be stopped at some point, and a pre-recorded explanation about the situation will be played which will be the discriminative stimulus to understand the expressive language of the character. Then, the child will be asked to identify the emotion of the character by selecting the most suitable emoji from the given. The seven basic emotions (with later added "neutral" emotion) introduced by Ekman according to facial action coding system (FACS) will be displayed through the emojis [5]. Fig. 2 shows an example view of emotion recognition component. To encourage engagement, differential reinforcement will be integrated in both components.

As the next step of the research, I am working on the ethics approval for a co-design process with children with ASD. This will inform the development of the system. I would appreciate feedback on the following issues in the doctoral consortium: Firstly, possible

features that can be incorporated with AR face masks to enhance the children's engagement in the conversation. Secondly, how possible it is to generalize when providing reinforcement to different children on different part of the spectrum.

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